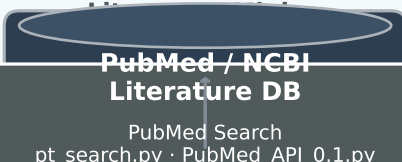


miRNA-seq Analysis Pipeline for IBD

Dataset PRJNA471862 · Ulcerative Colitis · End-to-End Workflow

DATA ACQUISITION



PubMed Search
pt_search.py · PubMed_API_0.1.py

SEQUENCING PROCESSING

00

Dataset Discovery & Metadata Curation

EDirect · SRA Toolkit [01_geo_ibd_curator.py, 02_bioproject_curator.py]

candidates_master.csv · runinfo.csv · dataset metadata

01

Raw Data Download

prefetch · fasterq-dump · pigz [03_download_runs.sh]

Raw reads · SRR*.fastq.gz (160 samples)

02

Quality Control

FastQC · MultiQC · miRTrace [04_qc_fastqc.sh, 06_qc_mirtrace.sh]

FastQC / MultiQC report · miRTrace QC report

03

miRNA Quantification

miRge3.0 · miRBase hsa [08_test_auto_mirge3_subfolders.sh, 09_single_end_runs_mirge3.sh]

miRNA.Counts.csv · miRNA.RPM.csv (per sample)

04

Count Merge & Filtering

pandas · mean RPM ≥ 100 filter [11_mrlna_cut_off.py, 12_deseq2_metadata.py]

Master_Filtered_Counts_DESeq2.csv · sample_metadata.csv

05

Differential Expression Analysis

DESeq2 · apeglm · VST · PCA / UMAP [13_run_dea.R]

DEA results · Default & Strict models

PCA · UMAP · Heatmap · Volcano

06

Target Prediction

multiMiR · miRTarBase · TarBase · miRecords [14_multimir_targets.R]

targets_validated.csv · targets_predicted.csv

07a

Pathway Enrichment

clusterProfiler · GO · KEGG [15_pathway_enrich.R]

GO_Enrichment.csv · KEGG_Enrichment.csv · dotplots

07b

Network Construction & Clinical Evaluation

igraph · clusterProfiler · pROC · Louvain [build_network.R, clinical_eval.R, 16_interactive_network.R]

miR-gene network · ROC/AUC · centrality scores

Interactive Dashboard

Plotly.js · vis.js · matplotlib [17_generate_report.py, 16_interactive_network.R]

- Legend
- Data Acquisition (00–01)
 - Quality Control (02)
 - Quantification & Processing (03–04)
 - Statistical Analysis (05)
 - Biological Interpretation (06–07)
 - Literature Mining (parallel)
 - Intermediate data / files
 - Data flow
 - Supplementary link